OAKLAND COMMUNITY COLLEGE®
ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

ROB 2140
Studio 5000 ADD ON INSTRUCTION LABS

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Studio 5000 ADD ON LABS

HMI SCREENS FOR THE APPLICATION PROGRAMS

Close **FactoryTalk View ME Studio** if open.

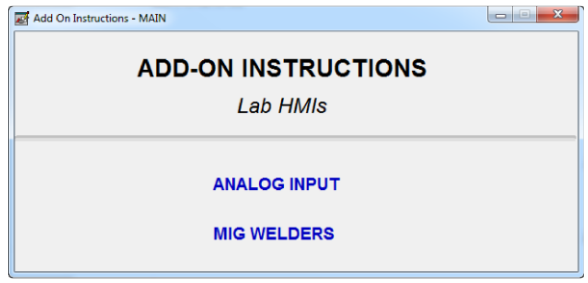
Open **FactoryTalk View ME Station**.

The HMI is a runtime file, see HMI Procedures, section Station Runtime Application to start the **Add_On_HMI** file.

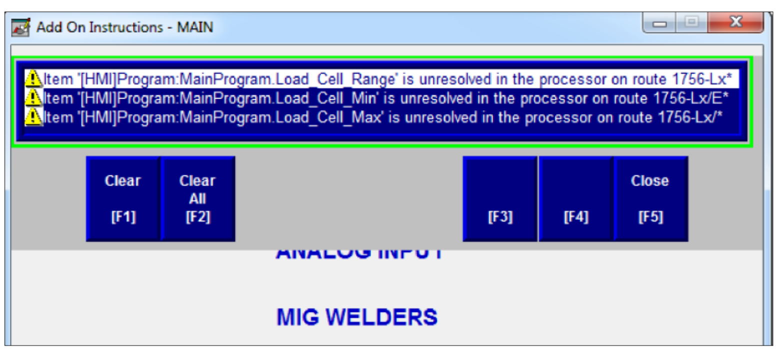
A Main menu appears:

1. **Do Not** open a application HMI screen until the RSLogix 5000 program has been downloaded and is in RUN mode.

2. **Return to the MAIN SCREEN** before changing the RSLogix 5000 processor to PROGRAM mode or OFFLINE.



After downloading the Studio 5000 program and placing in RUN mode, an alarm screen will appear because the processor went offline to download. Press F8.



LAB: ANALOG SCALING
PROCESS MODEL: Add_On_Analog

PROGRAM LOCATION:

Studio 5000 Folder
Session Folder
Add On Folder

FILE NAME: Add_On_Analog

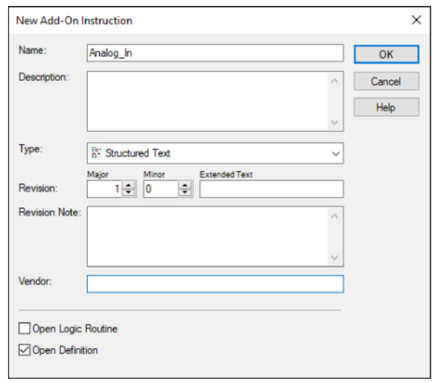
CREATE NEW ADD-ON INSTRUCTION AND NAME

In the Controller Organizer expand **Assets**, right click on **Add-On Instructions**. Select **New Add-On Instruction...** from the context menu.

Name: Analog_In

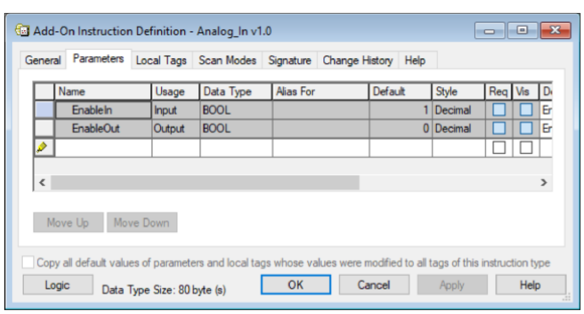
Type: Structured Text Diagram

Select the **OK** button.



CREATE ADD-ON PARAMETERS AND LOCAL TAG

In the **Add-On Instruction Definition** dialog, In the Controller Organizer, **Add-On Instructions**, expand the new **Analog_In** and double left click on **Parameter and Local Tags**.



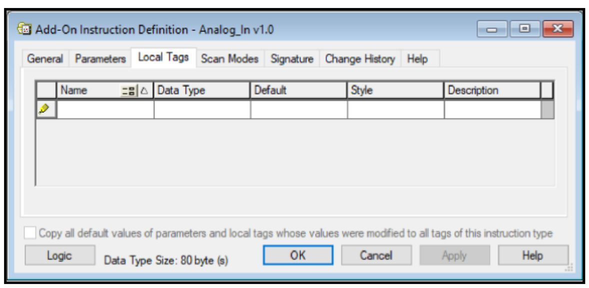
Add the Parameters listed on the next page.

Parameters:

Name	Usage	Data Type	Req/Vis	Descriptions
Raw_Max	Input	REAL	✓	Maximum Bit Value
Raw_Min	Input	REAL	✓	Minimum Bit Value
EU_Max	Input	REAL	✓	Engineering Unit Maximum
EU_Min	Input	REAL	✓	Engineering Unit Minimum
Analog_Ch	Input	REAL	✓	Analog Channel, Source
EU_Value	Output	REAL	✓	Scaled Engineering Unit Value

Select the **Apply** button when complete.

Select the **Local Tags** tab.



Add the local tag.

Local Tag:

Name	Data Type	Descriptions
Analog	SCALE	Data Type for Add On Instruction

Select the **OK** button when complete.

---

continue

CREATE ADD-ON INSTRUCTION LOGIC

In the Controller Organizer, **Add-On Instructions**, expand the new **Analog_In** and double left click on **Logic**.

Create a Structured Text Program using the Scale **SCL** instruction and add the Local Tag.

```
Analog.In := Analog_Ch;
Analog.InRawMax := Raw_Max;
Analog.InRawMin := Raw_MIn;
Analog.InEUMax := EU_Max;
Analog.InEUMin := EU_Min;
SCL(Analog);
EU_Value := Analog.Out;
```

Close when completed.

CREATE Controller AND MainProgram Parameters and Local Tags

CONTROLLER TAGS

In the Controller Organizer, select Controller Tags and add the tags for the networked raw (bit) value for the sensors.

Address	Controller Tag	Data Type	Descriptions
Process_Sensors:I.Data[0]	LOAD_CELL	DINT	Load Cell Raw (Bit) Input
Process_Sensors:I.Data[1]	FLOW_METER	DINT	Flow Meter Raw (Bit) Input

PARAMETER AND LOCAL TAGS

In the Controller Organizer, select MainProgram Parameters and Local Tags and add the tags **WEIGHT** and **FLOW** with the **Analog_in** Interface Data Type structure.

Parm/Local Tag	Data Type	Descriptions
WEIGHT	Analog_In	Add On Instruction Tag for Load Cell
FLOW	Analog_In	Add On Instruction Tag for Flow Meter

Tags are continued on the next page.

Parm/Local Tag	Data Type	Descriptions
Load_Cell_Max	DINT	Load Cell Engineering Unit Maximum Range
Load_Cell_MIn	DINT	Load Cell Engineering Unit Minimum Range
Load_Cell_Range	DINT	Load Cell Range Output in Engineering Units
FlowMeter_Min	DINT	Flow Meter Engineering Unit Maximum Rate
FlowMeter_Max	DINT	Flow Meter Engineering Unit Minimum Rate
FlowMeter_Rate	DINT	Flow Meter Rate Output in Engineering Units

NOTE: Tags for the Raw Bit Maximum and Minimum are not created, these values are hardware fixed and will be hardcoded in the instruction.

INFORMATION ONLY

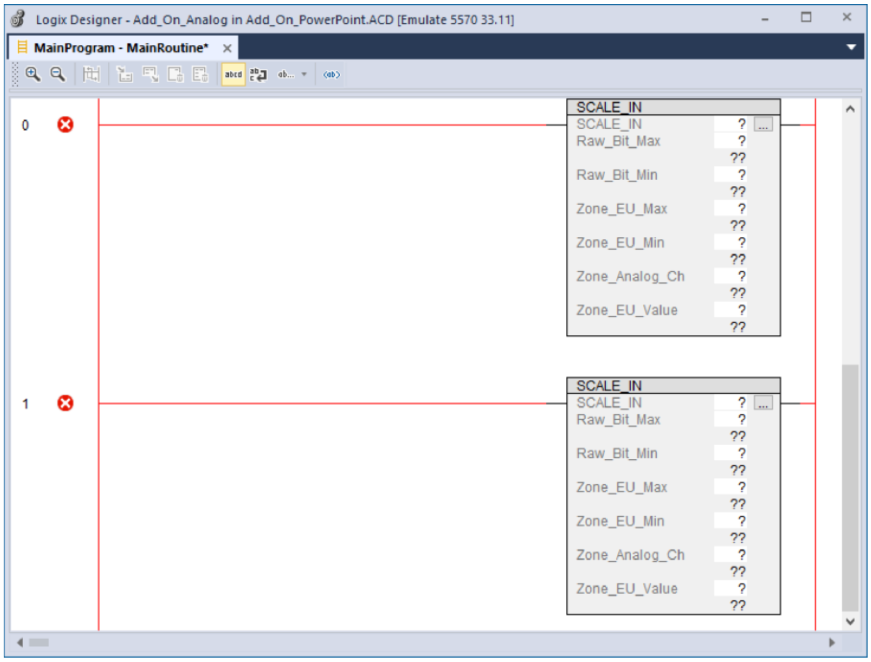
When **WEIGHT** or **FLOW** Tag Names are expanded the Parameters defined in the Add-on Instruction, ***created automatically when Tag Name is created***, are included in the data structure.

Example: WEIGHT.InRawMax

Tag Parameters (*Parameters are created when he WEIGHT or FLOW Tag is created*)	
tag.EnableIn	In to enable instruction (default is enabled)
tag.EnableOut	Out status of instruction (true when instruction In true)
tag.InRawMax	Raw bit maximum value to tag Analog data structure
tag.InRawMin	Raw bit minimum value to tag Analog data structure
tag.InEUMax	Eng. Unit maximum value to tag Analog data structure
tag.InEUMin	Eng. Unit minimum value to tag Analog data structure
tag.Out	Out from tag Analog data structure
tag.Analog_Ch	Analog Input, Raw (binary) data
tag.EU_Value	Scaled value in Engineering Units

MainRoutine LOGIC

In the MainRoutine, drag from the **Add-On** instruction tab toolbar the **Analog-In** instruction onto unconditional rungs.



Assigned the required tags for the Load Cell and Flow Meter.

Rung 0: Add On Instruction

Parameter	Tag/Constant
Analog_In	WEIGHT
Raw_Max	32767
Raw_Min	0
EU_Max	Load_Cell_Max
EU_Min	Load_Cell_Min
Analog_Ch	Load_Cell
EU_Value	Load_Cell_Range

Rung 1: Add On Instruction

Parameter	Tag/Constant
Analog_In	FLOW
Raw_Max	32767
Raw_Min	0
EU_Max	FlowMeter_Max
EU_Min	FlowMeter_Min
Analog_Ch	Flow_Meter
EU_Value	FlowMeter_Rate

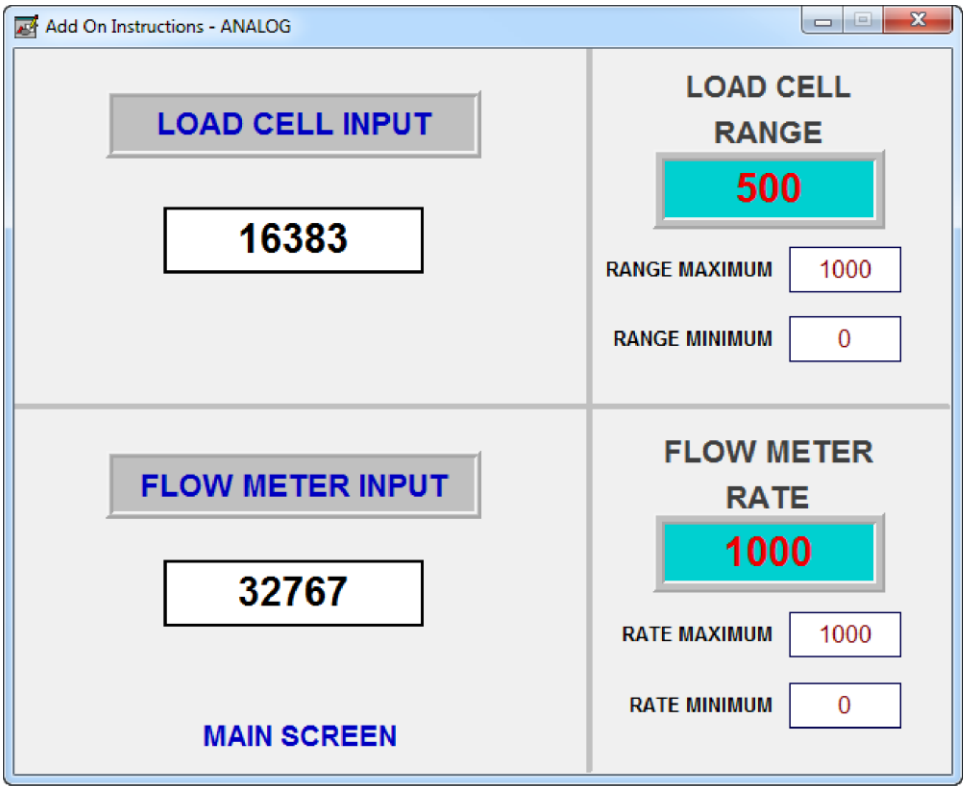
DOWNLOAD

Download and set mode to RUN.

SEE NEXT PAGE FOR HMI

HMI

For the **LOAD CELL** or **FLOW METER**, set the **RANGE** or **RATE MAXIMUM** and **MINIMUM** (Engineering Units) for the sensor by left clicking on the input field and entering a value.



For the **LOAD CELL** or **FLOW METER**, set the Bit Value (Raw) **INPUT** for the sensor by left clicking on the **NUMERIC FIELD** and entering a value.

Left click on the **LOAD CELL INPUT** or **FLOW METER INPUT** button.

The scaled **RANGE** or **RATE** is displayed.

LAB: MIG STATION
PROCESS MODEL: Add_On_MIG

PROGRAM LOCATION:
- **Studio 5000** Folder
- **Session** Folder
- **Add On** Folder

FILE NAME: Add_On_MIG

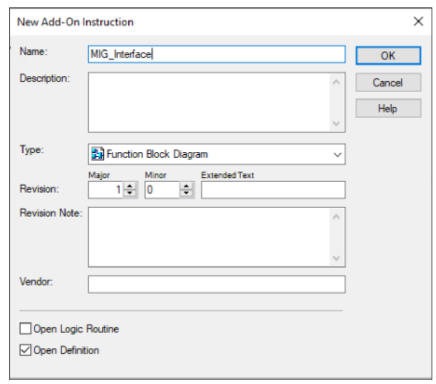
CREATE NEW ADD-ON INSTRUCTION AND NAME

In the Controller Organizer expand **Assets**, right click on **Add-On Instructions**. Select **New Add-On Instruction...** from the context menu.

Name: MIG_Interface

Type: Function Block Diagram

Select the **OK** button.



CREATE ADD-ON PARAMETERS AND LOCAL TAGS

In the **Add-On Instruction Definition** dialog, select the **Parameters** tab.

PARAMETERS:

Name	Usage	Data Type	Req/Vis	Descriptions
Arc_Start	Input	BOOL	✓	Begin MIG Welding
Arc_End	Input	BOOL	✓	MIG Welding Finished
Wire_Feed	Output	BOOL	✓	Energize Filler Wire Feeder
Gas_Flow	Output	BOOL	✓	Open Shielding Gas Valve
Arc_On	Output	BOOL	✓	MIG Arc Established

Parameters are continued on next page.

Name	Usage	Data Type	Req/Vis	Descriptions
Pre_Weld_Delay	Input	DINT	✓	Gas and Wire Preflow
Fault_Arc	Input	BOOL	✓	Loss of MIG Welding Arc
Fault_Wire	Input	BOOL	✓	Wire feed not detected

Select the **Apply** button when complete.

Select the **OK** button when complete.

CREATE ADD-ON INSTRUCTION LOGIC

In the Controller Organizer, **Add-On Instructions**, expand the new **MIG_Interface** and double left click on **Logic**.

When the Function Block instructions are dragged into the program the Local Tags ***are created automatically***.

Local Tag	Data Type	Descriptions
BOR_01	FBD_BOOLEAN_OR	OR for Arc Start/Timer Enabled
BNOT_01	FBD_BOOLEAN_NOT	Not for Arc End
BAND_01	FBD_BOOLEAN_AND	AND for Arc Start, Arc End, Timer Enable, and Faults
TONR_01	FBD_TIMER	Timer for Pre-Weld

Create the following Function Block Program.

See the diagram on the next page.

A set **Arc_Start** begins the welding process.

The TONR instruction's Enabled holds the circuit after Arc_Start is released.

The TONR instruction's PRE is based on the **Pre_Weld_Delay** setting.

While the timer is elapsing, the **Wire_Feed** and **Gas_Flow** is set.

All conditions must be set:

Timer **Arc_Start** timer enabled.
No **Fault_Arc**, signal is set.
No **Fault_Wire**, signal is set.

When the timer is complete the **Arc_On** is set.

All conditions must be set:

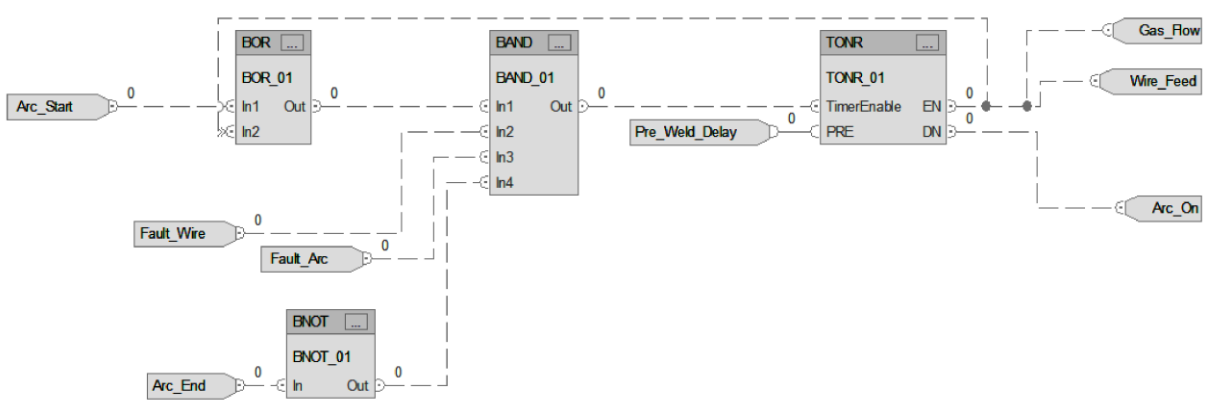
Timer **Arc_Start** timer is done.
No **Fault_Arc**, signal is set.
No **Fault_Wire**, signal is set.

Note: The **Wire_Feed** and **Gas_Flow** remain set when the **Arc_On** is set.

A Set **Arc_End** turns Off **Wire_Feed**, **Gas_Flow, and Arc_On**.

Note: The **Fault_Arc** and **Fault_Wire** signal is normally high (set) when there is ***no fault***. The **Arc_Start** and **Arc_End** are normally low (reset) signals.

Use documents Zoom feature if needed to view.



continue

CREATE MainProgram Parameters and Local Tags

In the Controller Organizer, select the MainProgram Parameters and Local Tags and add the tags **Welder1** and **Welder1** with the **MIG_Interface** Data Type structure.

Parm/Local Tag	Data Type	Descriptions
Welder1	MIG_Interface	MIG Welding Station No. 1
Welder2	MIG_Interface	MIG Welding Station No. 2

USE OF ADDRESSES INSTEAD OF TAGS FOR ADD-ON INSTRUCTION

In the program below, the actual I/O addresses become an "Alias" Tag to the Add On Instruction's parameters.

In the previous Analog Add On, alias Local and Parameter Tags were assigned to the Add On Instruction's parameters.

In the Analog Add On, the data has been "localized" by assigning Program tags to the Add instruction's parameters.

In this program, the MIG Add On parameters are assigned physical I/O, which are by default Global, the parameter Tags are then "localized" by the Add On structures Program defined tag.
Notice the Parameter's Tag name can be assigned to other logic instructions in the program. Example, the interlocking for Arc or Wire faults.

continue

MainRoutine LOGIC

In the MainRoutine, drag from the Add-On instruction tab toolbar, the MIG_Interface instruction onto unconditional rungs, and assigned the required tags and addresses.

Rung 0: Add On Instruction

Parameter	Tag/Address
MIG_Interface	Welder1
Arc_Start	Local:3:I.Data.0
Arc_End	Local:3:I.Data.1
Wire_Feed	Local:4:O.Data.0
Gas_Flow	Local:4:O.Data.1
Arc_On	Local:4:O.Data.2
Pre_Weld_Delay	Welder1.Pre_Weld_Delay
Fault_Arc	Local:3:I.Data.2
Fault_Wire	Local:3:I.Data.3

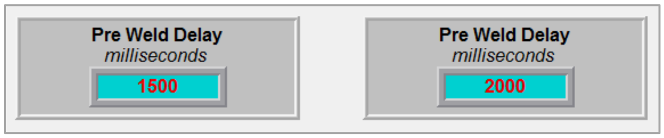
Rung 1: Add On Instruction

Parameter	Tag/Address
MIG_Interface	Welder2
Arc_Start	Local:3:I.Data.4
Arc_end	Local:3:I.Data.5
Wire_Feed	Local:4:O.Data.3
Gas_Flow	Local:4:O.Data.4
Arc_On	Local:4:O.Data.5
Pre_Weld_Delay	Welder2.Pre_Weld_Delay
Fault_Arc	Local:3:I.Data.6
Fault_Wire	Local:3:I.Data.7

DOWNLOAD

Download and set mode to RUN.

Enter a Welder*number*.Pre_Weld_Delay for Welder 1 and Welder 2 in either the **Parameters and Local Tags** and **Monitor Tags** tab or the **Watch Window**.



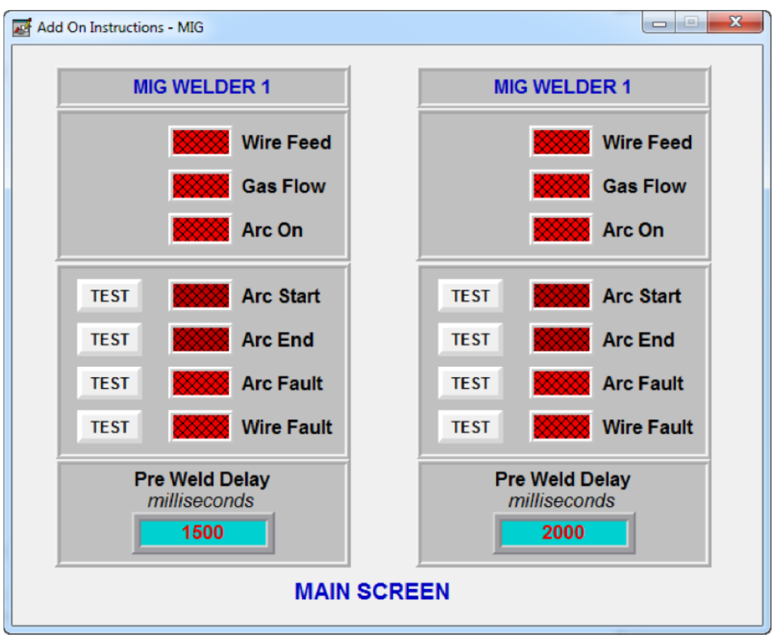
HMI

Objects for MIG Welder 1 and MIG Welder 2.

Press the button TEST for **Arc Start**, the **Wire Feed** and **Gas Flow** turns On and after the delay **Arc On** is energized.

Press the button TEST for **Arc End** below the panel to turn Off the welder's outputs.

Press the button TEST for **Arc Start**, while welding, press a TEST button for either the **Arc** or **Wire Fault** to induce the fault.



Note: Arc Fault and Wire Fault are normal "High" (Logic 1) signal when no fault exists. A fault reset the signal to a "Low" (Logic 0) status.

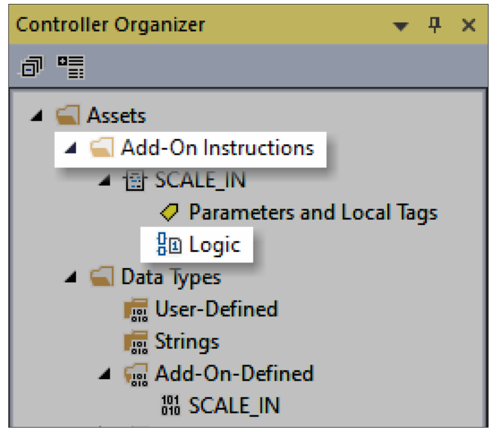
TO EDIT ADD-ON INSTRUCTION LOGIC

Expand **Assets** and **Add-On Instructions**.

Expand the named Add-On instruction.

Double left click on **Logic** to open the editor.

Logic Editor opens.



TO EDIT ADD-ON INSTRUCTION PARAMETERS AND LOCAL TAGS

Expand **Data Types**.

Expand **Add-On_Defined**.

Double left click on the named Add-On to open the dialog.